American Computer Science League

2020-2021 ● Contest 1: Solutions ● Elementary Division

1. Computer Number Systems

$41375_8 = 100\ 001\ 011\ 111\ 101_2 = 100\ 0010\ 1111\ 1101_2 = 42FD_{16}$

1. $42FD_{16}$ (A)

2. Computer Number Systems

$100101110_2 = 456_8$ or $12E_{16}$ $\quad 457_8 = 100101111_2 = 12F_{16}$
$12D_{16} = 100101101_2 = 455_8$

The smallest of these in base 8 is 455 and the smallest in base 16 is 12D.

2. $12D_{16}$ (C)

3. Computer Number Systems

$$\begin{array}{r} 6023_8 \\ +\ \underline{757_8} \\ 7002_8 \end{array} \qquad \begin{array}{r} 7002_8 \\ +\ \underline{1757_8} \\ 10761_8 \end{array}$$

Remember that you must carry 8, not 10 when adding in base 8.

3. 10761_8 (B)

4. Computer Number Systems

Counting from 16 to 35 is as follows:

16 = 10000	21 = 10101	26 = 11010	31 = 11111
17 = 10001	22 = 10110	27 = 11011	32 = 100000
18 = 10010	23 = 10111	28 = 11100	33 = 100001
19 = 10011	24 = 11000	29 = 11101	34 = 100010
20 = 10100	25 = 11001	31 = 11110	35 = 100011

The number of 1's from 16 to 25 is 25 and the number of 1's from 26 to 35 are 31. The difference is 6.

4. 6 (C)

5. Computer Number Systems

All colors are represented by RGB values which are $25A8F6_{16}$ and $C2D9AB_{16}$ so the blue components are $F6_{16}$ and AB_{16}. The

5. $4B_{16}$ (A)

difference is $F6_{16}$ - AB_{16} = $4B_{16}$.	